

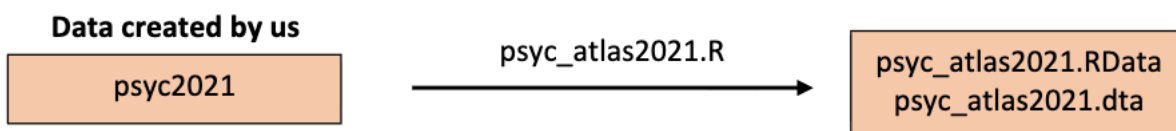
psyc_atlas2021

Version: 1 (February 10, 2023)

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The dataset `psyc_atlas2021` contains information on the date of the first diagnosis of 27 pre-defined mental disorders and broad categories of mental disorders (Table 1). These disorders are diagnosed in Danish psychiatric departments until August 24, 2022 for all individuals living in Denmark between January 1, 1986, and December 31, 2021. There is one observation for each individual with a specific date for each of the diagnoses.

Code: `psyc_atlas2021.R` (R code)
Output: `psyc_atlas2021.RData` (R file)
`psyc_atlas2021.dta` (Stata file)



The dataset `psyc_atlas2021` is created based on register `psyc2021`, which includes information on all diagnoses of mental disorders in Danish psychiatric departments. Specific details and R code for register `psyc2021` are available on OSF (<https://osf.io/zhfyp/>). In brief, we have created register `psyc2021` based on the Danish Psychiatric Central Research Register,¹ which contains information on admissions to psychiatric inpatient facilities since April 1969 and visits to outpatient psychiatric departments and emergency departments since January 1, 1995. Although the register includes data from 1969, it was on January 1, 1970 that all psychiatric departments were committed to reporting the data; thus, year 1970 is the first year with valid data. The diagnostic system used is the Danish modification of the *International Classification of Diseases, Eighth Revision* (ICD-8) from 1969 to 1993, and *Tenth Revision* (ICD-10) from 1994 onwards. The register is available through different specific registers from the [Danish Health Data Authority](#) ([Sundhedsdatastyrelsen; SDS](#)). We considered mental disorders classified as main diagnoses (*hoveddiagnoser/aktionsdiagnoser*; type 'A'), auxiliary diagnoses (*bidiagnoser*; type 'B'), basic diagnoses (*grundmorbus*; type 'G'), and associated diagnoses (*tillægsdiagnoser*; type '+'). The dataset does not include referral diagnoses (*henvisningsdiagnose*; type 'H') or diagnoses linked to a hospital contact in which the main diagnosis is related to healthy individuals (i.e., ICD-8 code Y719; ICD-10 codes Z03.2, Z03.8, Z03.9, Z76.3).

The dataset `psyc_atlas2021` contains the date of first diagnosis for each of the disorders listed in Table 1. The final dataset contains 1,055,820 observations from 1,055,820 unique identifiers with the following variables:

- **pnr**: Unique personal identifier. All individuals born or immigrating to Denmark receive a unique personal registration number (CPR number) containing 10 digits. Based on different digits' positions, the CPR number contains information on the date of birth and sex. The [PNR number](#) is a de-identified version of the CPR number

made by DST that does not include personal information but allows linkage between different registers based on this unique identifier.

- **sex:** Sex based on DST's variable [koen](#) (1: male; 2: female), indicating the legal sex of the individual based on information from the CPR number. For more information, please look at documentation for register [population1986_2021](#) (available on OSF; <https://osf.io/zhfyp/>).
- **birth_d:** Date of birth based on DST's variable [foed dag](#).
- **Dxx, D00, ..., D91:** Date of the first diagnosis of disorders listed in Table 1. For each disorder, we considered only diagnoses happening after the earliest possible age of onset as listed in Table 1.

Table 1. Classification of mental disorders into 27 specific mental disorders or broad categories of disorders. This classification is slightly adapted from a previous publication using Danish registers.²

	Mental disorders	ICD-10	ICD-8 equivalency	Min age
Dxx	Any mental disorder	F00-F99	290-315	1
D00	Organic, including symptomatic, mental disorders	F00-F09	290.09, 290.10, 290.11, 290.18, 290.19, 292.x9, 293.x9, 294.x9 (except 294.39), 309.x9	35
D01	<i>Dementia in Alzheimer disease</i>	F00	290.09, 290.10, 290.19	35
D02	<i>Vascular dementia</i>	F01	293.09, 293.19	35
D10	Mental and behavioral disorders due to psychoactive substance use	F10-F19	291.x9, 294.39, 303.x9, 303.20, 303.28, 303.90, 304.x9	10
D11	<i>Mental and behavioral disorders due to alcohol use</i>	F10	291.x9, 303.x9, 303.20, 303.28, 303.90	10
D12	<i>Mental and behavioral disorders due to cannabis use</i>	F12	304.59	10
D20	Schizophrenia and related disorders	F20-F29	295.x9, 296.89, 297.x9, 298.29-298.99, 299.04, 299.05, 299.09, 301.83	10
D21	<i>Schizophrenia</i>	F20	295.x9 (excluding 295.79)	10
D22	<i>Schizoaffective disorder</i>	F25	295.79, 296.89	10
D30	Mood disorders	F30-F39	296.x9 (excluding 296.89), 298.09, 298.19, 300.49, 301.19	10
D31	<i>Bipolar disorders</i>	F30-F31	296.19, 296.39, 298.19	10
D32	<i>Recurrent depressive disorder</i>	F33	296.09, 296.29, 298.09, 300.49 ^a	10
D33	<i>Single and recurrent depressive disorder</i>	F32-F33	296.09, 296.29, 298.09, 300.49	10
D41	Neurotic, stress-related, and somatoform disorders	F40-F48	300.x9 (excluding 300.49), 305.x9, 305.68, 307.99	5
D42	<i>Obsessive-compulsive disorder</i>	F42	300.39	5
D51	Eating disorders	F50	306.50, 306.58, 306.59	1
D52	<i>Anorexia nervosa</i>	F50.0	306.50	1
D61	Specific personality disorders	F60	301.x9 (excluding 301.19), 301.80, 301.81, 301.82, 301.84	10
D62	<i>Borderline-type</i>	F60.31	301.84	10
D63	<i>Antisocial personality disorder</i>	F60.2	301.79, 301.82	10
D70	Intellectual disabilities	F70-F79	311.xx, 312.xx, 313.xx, 314.xx, 315.xx	1
D81	Pervasive developmental disorders	F84	299.00, 299.01, 299.02, 299.03	1
D82	<i>Childhood autism</i>	F84.0	299.00	1
D91	Behavioral and emotional disorders with onset usually occurring in childhood and adolescence	F90-F98	306.x9 (except 306.59), 308.0x	1
D92	<i>Hyperkinetic disorder</i>	F90	308.01	1

^aOnset was defined as the admission that occurred at least 8 weeks after last discharge with the ICD-8 codes.

1. Mors O, Perto GP, Mortensen PB. The Danish Psychiatric Central Research Register. *Scand J Public Health*. 2011;39(7 Suppl):54-57. doi:10.1177/1403494810395825
2. Pedersen CB, Mors O, Bertelsen A, et al. A comprehensive nationwide study of the incidence rate and lifetime risk for treated mental disorders. *JAMA Psychiatry*. 2014;71(5):573-581. doi:10.1001/jamapsychiatry.2014.16

